



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2023

INFORMATION AND COMMUNICATIONS TECHNOLOGY (N540)

Candidates answer on the Question Paper.

TIME: 1 HOUR 45 MINUTES

INSTRUCTIONS TO CANDIDATES

1. Write your index number in the space provided above.
2. Write in dark blue or black ink. Do not use correction fluid.
3. You may use a soft pencil for any diagram, graph or rough working.
4. Any rough working should be done in this booklet.
5. Do NOT use calculators.
6. Answer **ALL** questions.
7. This document consists of **11** questions printed on **22** pages, numbered **2** to **23**.
8. The number of marks is given in brackets [] at the end of each question or part question.
9. The total marks for this paper is **100**.

For Examiners' use													
Question No.	1	2	3	4	5	6	7	8	9	10	11	Total	Signature
Examiner													
Team Leader													
CE/ACE													

SECTION A

Question 1 (15 marks)

Circle the letter corresponding to the correct answer. Each item carries one mark.

(a) Which device is used to produce a **hard copy** of a document?

A



Mouse

B



Monitor

C



Scanner

D



Printer

(b) Which type of **microcomputer** is shown in the picture?

A Desktop computer

B Smartphone

C Tablet PC

D Laptop computer

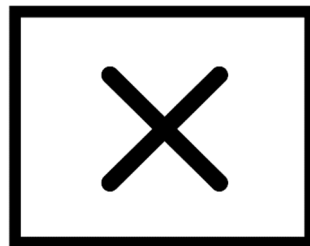


(c) Which **device** is shown in the picture below?



- A Speaker
- B Plotter
- C Headphone
- D Video projector

(d) Which **button** is shown below?



- A Close
- B Minimise
- C Maximise
- D Restore

- (e) Which **secondary storage device** is shown in the picture below?



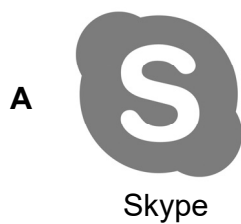
- A Compact disc
 - B Memory card
 - C Pendrive
 - D Hard disc drive
- (f) Study the date given below.

29 **March** 2022

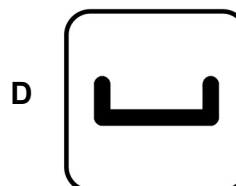
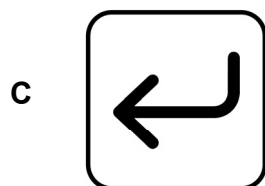
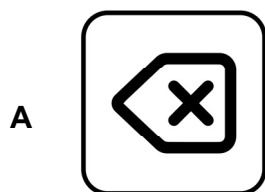
The word 'March' is formatted in

- A italics
 - B underline
 - C bold and italics
 - D bold and underline
- (g) **Backup** means to an original document.
- A delete
 - B modify
 - C keep a copy of
 - D print a copy of

(h) Which of the following is a **search engine**?



(i) Which one is the **Enter Key** on a keyboard?

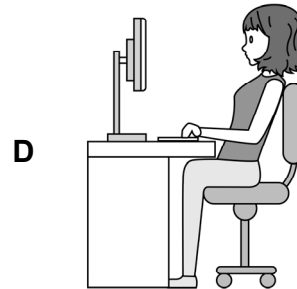
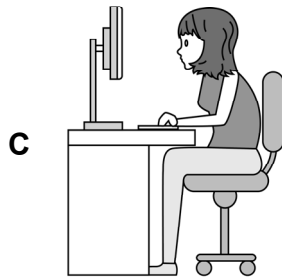
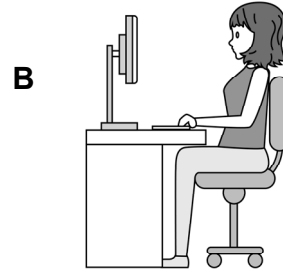


(j) Which **speech balloon** is shown below?

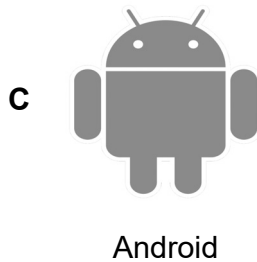
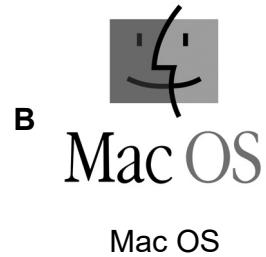


- A Sound balloon
- B Thought balloon
- C Secret balloon
- D General Speech balloon

(k) Which diagram shows the correct **sitting posture** at a computer?



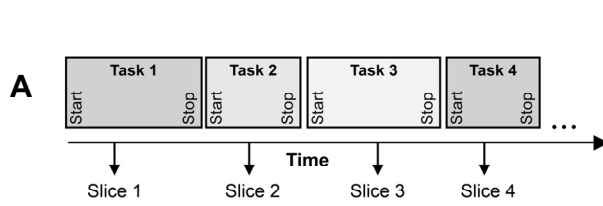
(l) Which one of the following is an **operating system** used with **smartphones**?



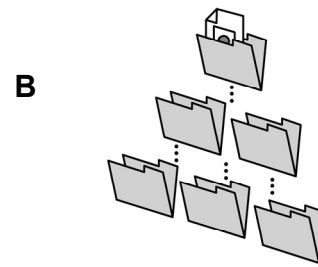
(m) Which one of the following is an example of a **computer crime**?

- A** Playing online games
- B** Printing a document
- C** Writing an email to a friend
- D** Hacking a computer

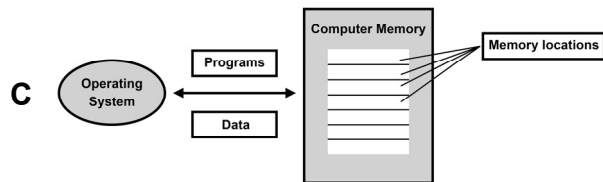
(n) Which one of the following helps to prevent **unauthorised access** to a computer?



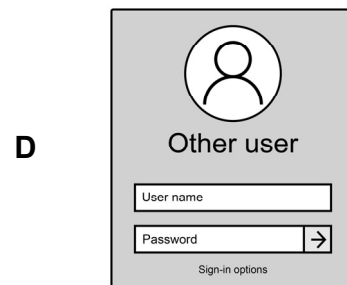
Processor Management



File Management

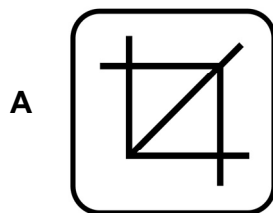


Memory Management

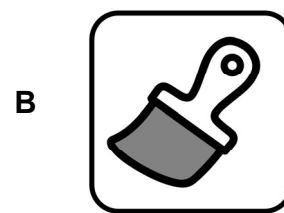


System Security Management

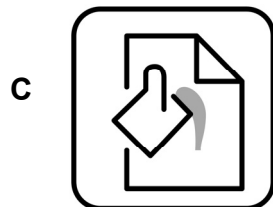
(o) Which tab would you use to remove **unwanted parts** of a picture?



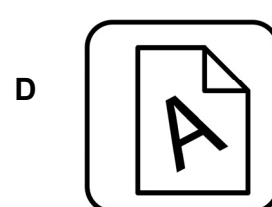
Crop



Format Painter



Page Colour

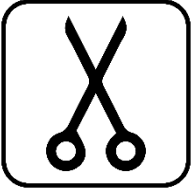


Watermark

Question 2 (10 marks)

- (a) Name the following **icons** used in Word Processing. Choose from the list of words given below.

Print	WordArt	Cut
Copy	Paste	Save

[5]

(b) Fill in the blanks with the **correct term** from the given list.

mailing list	power	eye strain
network	Hertz	volume

- (i) If a computer freezes, you should switch off the
- (ii) If you cannot hear the sound, check the level.
- (iii) The speed of a computer is measured in
- (iv) Staring at a computer screen for long hours can lead to
- (v) The contains information like names, addresses and phone numbers.

[5]

Question 3 (10 marks)

Tick (✓) **True** or **False** next to each of the statements below. An example is given.

		True	False
(a)	A mouse is an output device.		✓
(b)	Wifi means wireless fidelity.		
(c)	Page border adds colour to a page.		
(d)	Food and drinks are allowed in a computer lab.		
(e)	A handout is a hardcopy of PowerPoint slides.		
(f)	Online forums are websites used for debates.		
(g)	Any information collected should be kept safe.		
(h)	It is ethical to use someone else's password.		
(i)	A spreadsheet is used to create charts.		
(j)	A paper jam prevents a printer from working properly.		
(k)	One byte is equal to 4 bits.		

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Question 4 (10 marks)

- (a) Classify the **hardware devices** given below as Input, Output or Storage devices. Tick (✓) the appropriate column. An example is given.

		Input Device	Output Device	Storage Device
(i)	 Monitor		✓	
(ii)	 Digital Camera			
(iii)	 Joystick			
(iv)	 Blu-ray Disc			
(v)	 3D Printer			

[4]

- (b) Match each term in **Column A** to its correct definition or description in **Column B**. Write your answers in the **answer grid** provided below. An example is given.

Column A	
(i)	Terminal
(ii)	Software
(iii)	Keyboard
(iv)	Repetitive Strain Injury
(v)	Virus
(vi)	Intranet
(vii)	Temperature Sensor

Column B	
1	is used to enter text.
2	causes pain in wrists and fingers.
3	is a private network.
4	is the name given to computer programs.
5	is a computer connected to a network.
6	is used in automatic cooling systems.
7	connects networks across the world.
8	is a software that can damage a computer.

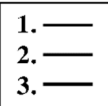
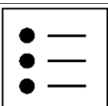
(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
5						

[6]

Question 5 (5 marks)

(a) Using the words given below, name the following icons. An example is given.

Centre	Bullets	Justify	Align left	Numbering
--------	---------	---------	------------	-----------

	 Justify
	
	
	

[3]

(b) John has accidentally deleted a file. He wants to restore it from the **recycle bin**.



Using 1, 2 and 3, put the following steps in the **correct order** to help John restore the file.

Step	Order
Click on restore.	
Open recycle bin.	
Right click on file.	

[2]

SECTION B

Question 6 (5 marks)

The specifications for a computer system are provided below.

About



Windows 10

PC name IDG-JIM-WORK-PC

[Rename this PC](#)

Organisation

Manage or disconnect from work or school

Edition	Windows 10 Pro
Version	1703
OS Build	15064.674
Product ID	
Processor	Intel(R) Core(TM) i5-6600K CPU @ 3.50GHz 3.50 GHz
Installed RAM	8.00 GB
System type	64-bit operating system x64-based processor
Pen and touch	No pen or touch input is available for this display

- (a) Tick (✓) the correct **processor model** used in this computer system.

Intel Core i3 ☐ Intel Core i5 ☐ AMD Pro ☐

[1]

- (b) What is the **speed** of the processor?

..... [1]

(c) What is the **memory size** of the computer system?

..... [1]

(d) Which **edition** of the operating system is used in this computer system?

.....[1]

(e) Give **one function** of an operating system.

..... [1]

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Question 7 (8 marks)

(a) The following sentences refer to the features of Microsoft Word.

Fill in each blank with the **correct term** from the list given below.

view	layout	page break	section break
------	--------	------------	---------------

(i) A is used to separate contents between pages.

(ii) The tab is used to change the orientation of a page from portrait to landscape.

(iii) A splits pages of a document into parts allowing one to use different formatting on each page.

[3]

- (b) A table of contents of a textbook is shown below.

Contents		
Chapter 1.0	Introduction	2
Chapter 2.0	Project Requirements	7
Chapter 3.0	The History of Grammar	12
Chapter 4.0	Parts of Speech	18
4.1	Nouns	21
4.2	Verbs	24
4.3	Adjectives	27
4.4	Adverbs	31
Chapter 5.0	Sentences	35
5.1	Complex Sentences	39
5.2	Compound Sentences	45
Chapter 6.0	Paragraphs	51
6.1	Descriptive	56
6.2	Expository	60
6.3	Narrative	65
6.4	Persuasive	71
Chapter 7.0	Conclusion	76

- (i) Tick (✓) the correct **tab** which is used to insert the table of contents?

View ☐ References ☐ Mailings ☐ [1]

- (ii) On which page does **Chapter 3** start?

..... [1]

- (iii) How many chapters are there in this textbook?

..... [1]

- (iv) What is the name of the chapter that starts on **page 51**?

..... [1]

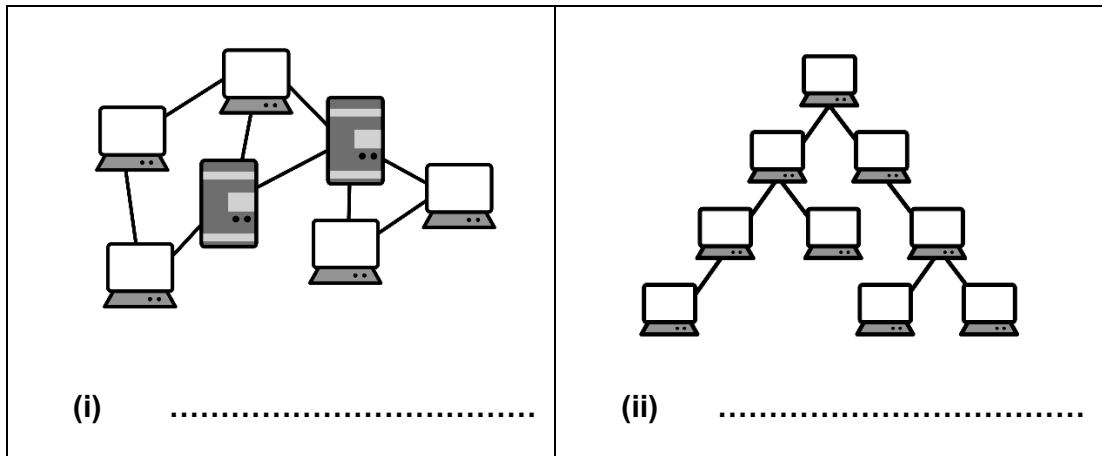
- (v) What is the name of the **sub-chapter 6.3**?

..... [1]

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Question 8 (9 marks)

- (a) Name each **type of network topology** shown in the diagram below.



[2]

- (b) Tina uses **social networking** to communicate with her friends.

- (i) Explain the term **social networking**.

.....
 [1]

- (ii) Give two examples of social networking websites.

1.
 2. [2]

- (iii) Give two ways to **communicate** on a social networking website.

1.
 2. [2]

- (c) Give **two** safe practices that should be adopted when using a social networking website.

1.
 2. [2]

Question 9 (10 marks)

The spreadsheet below shows the data about the sales of JMS Candy Shop.

Refer to the spreadsheet and answer the questions that follow.

	A	B	C	D	E	F
1	JMS Candy Shop					
2	Item	Cost Price	Selling Price	Profit Per Item	Number in stock	Reorder stock
3	Bounty	26.50	28.50	2.00	167	NO
4	Twix	18.00	22.00	4.00	145	NO
5	Galaxy	22.50	25.00	2.50	32	YES
6	KitKat	32.00	32.50	0.50	159	NO
7	Mars	15.50	19.50	4.00	12	YES
8						
9					Items to be reordered	

(a) Write down which **cell** contains the following information:

(i) Cost Price of Twix:

(ii) Selling Price of Bounty:

[2]

(b) Tick (✓) the correct formula that has been used in cell **D3** to calculate the **Profit Per Item** on **Bounty**.

= B3 / C3 ☐ = MIN (B3:C3) ☐ = C3 – B3 ☐

[1]

(c) Write the formula that can be used in cell D8 to calculate the total **Profit Per Item**.

..... [2]

(d) When the number of a particular item is **less than 50**, the shop needs to reorder the stock.

Complete the following formula to be used in **cell F3** to find out whether there is a need to reorder stock or not.

= IF (E3 < 50,,,)

[2]

- (e) Conditional formatting is applied to cell E3:E7 as follows:

Name the **cells** that will be formatted.

..... [2]

- (f) The formula `= COUNTIF (F3:F7,"YES")` is used in cell F9.

What will be the output?

..... [1]

Question 10 (8 marks)

- (a) State whether each of the following statements on databases is **True** or **False**.
An example is given.

(i) Retrieving information from a computerised database is difficult. **False**

(ii) A primary key is a field which uniquely identifies a record.

(iii) Each row of a database table represents a record.

[2]

- (b) The table below shows the computerised database that stores details on the departure of flights.

The **FLIGHTS** table is shown below.

FLIGHT_ID	FLIGHT_NAME	DEPARTURE	LOCATION	DESTINATION
MK-054	Air Mauritius	04:00	Mumbai	Mauritius
MK-001	Air Mauritius	08:00	Mauritius	Rodrigues
AS-102	Air Austral	09:00	St Denis	Mauritius
SP-205	Singapore Airlines	23:00	Mauritius	Singapore
EM-234	Fly Emirates	23:00	Dubai	Mauritius
BA-458	British Airways	07:00	Mauritius	London

- (i) How many fields are there in the FLIGHTS table?

..... [1]

- (ii) The following search condition was entered using a **query by example** grid.

Field	FLIGHT_ID	DEPARTURE
Table	FLIGHTS	FLIGHTS
Sort	ASCENDING	
Show	☒	☐
Criteria		= 23:00
Or		

Show what the **output** would be.

.....

 [3]

- (iii) Complete the **query by example** grid below to select and show the **name of flights** located in **Mauritius**.

Field	FLIGHT_NAME	LOCATION
Table	FLIGHTS	FLIGHTS
Sort		
Show	☐	☐
Criteria		
Or		

[2]

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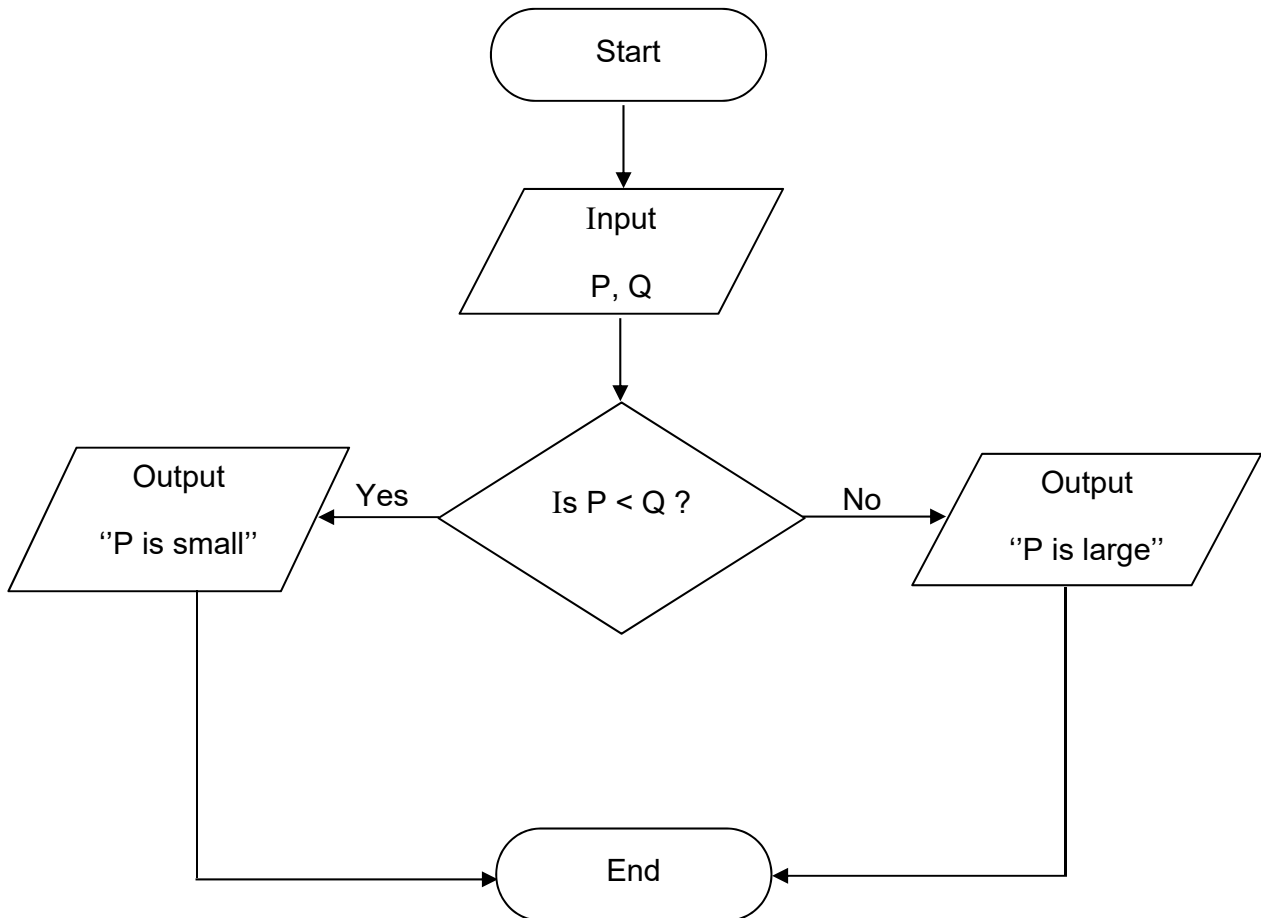
Question 11 (10 marks)

(a) Match the term in **Column 1** to the description in **Column 2**.

Column 1		Column 2
Loop	●	● Statements executed in the order in which they occur.
Selection	●	● Statements are executed multiple times.
Sequence	●	● Statements that are not run in a program.
		● Statements executed only if a certain condition is met.

[2]

(b) Study the flowchart below carefully.



For the sets of input given below, **dry run** the flowchart.

P	Q	Output
18	6	
6	14	
12	8	

[3]

(c) Read the following Python commands

```

Line 1 >>> totalprice = 2010
Line 2 >>> if totalprice > 2000:
Line 3 >>> print('You get a discount')
Line 4 >>> else:
Line 5 >>> print('You do not get a discount')
    
```

(i) What will be the output if the above Python code is run?

..... [2]

(ii) At **Line 1**, rewrite the command that will allow a user to enter the value for total price from the keyboard.

.....
 [3]

END OF PAPER

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